

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/ CSE/ PEC-CS502C/ 2024-25
2024
ARTIFICIAL INTELLIGENCE

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer **all** questions

5x2=10

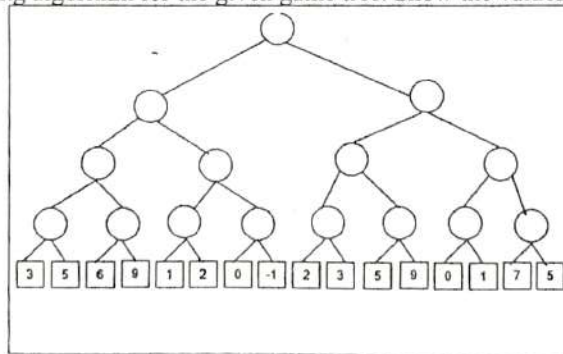
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|----|--|---|
| 1. | What is rational agent? Write down the basis of Rational agent. | 2 |
| 2. | Define Heuristic cost and Generation cost. | 2 |
| 3. | Write down the differences between goal-based agent and utility-based agent. | 2 |
| 4. | Write down the features of Adversarial search. | 2 |
| 5. | Write down the differences between Fuzzy set and Crisp set. | 2 |

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any **four** questions

4x15 = 60

- | | | |
|----|--|-----|
| 6. | a) What is AI agent? Explain the different components of AI agent. | 1+3 |
| | b) What is the importance of Environment in AI? Explain different Features of AI environment. | 1+6 |
| | c) Explain how Learning agent works with proper diagram. | 4 |
| 7. | a) Explain informed search and blind search techniques. Give examples. | 6 |
| | b) Write down the A* algorithm and explain with suitable example. | 5 |
| | c) Draw a neat state space diagram of Hill Climbing Algorithm and Explain the limitations. | 4 |
| 8. | a) Using Constraint Satisfaction Procedure, solve the crypt-arithmetic problem: SEND + MORE = MONEY | 7 |
| | b) Apply alpha-beta pruning algorithm for the given game tree. Show the values of Alpha and Beta at each step. | 8 |



- | | | |
|-----|--|-------|
| 9. | a) Draw a Semantic Network representing the following knowledge: (i) Tomy is a cat (ii) Tomy is Brown colored (iii) Tomy is a mammal (iv) Tomy is owned by Rishi (v) All mammals are animal | 3 |
| | b) Given two jugs, one 4 liters and the other 3 liters with no marking in them. Also given a water supply with a large storage. Write down all production rules that are used to measure 2 liters of water using these two jugs. | 7 |
| | c) What is knowledge? How it differs from Intelligence? Explain the different types of knowledge? | 1+1+3 |
| 10. | a) What is the importance of Learning in AI? Write down the differences between Supervised learning, Unsupervised learning and Reinforcement learning. | 2+6 |
| | b) Give some real world applications of AI. Explain the different steps of natural language processing. | 2+5 |

11. a) What do you mean by reasoning in AI. Explain different types of reasoning with suitable examples. ✓ 2+5
b) Explain the differences between forward chaining and backward chaining with examples. ✓ 4
c) What is an Expert System in AI? Explain the key components of an Expert System. ✓ 1+3
12. a) write down the truth tables for Conjunction and Implication. How uncertainty can be handled in logic based knowledge representation? What are the difficulties in NLP? Give suitable examples. 3+1+3
b) Represent the following predicates in First Order Predicate Logic: (i) It is crime for Americans to sell the weapon to the enemy of America. (ii) Country "X" is an enemy of America (iii) "X" has some missiles. (iv) All missiles were sold to "X" by Amran (v) Amran is an American (vi) missiles is a weapon 6+2
c) Using Forward chaining proof that Amran is criminal.
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JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/CSE/PCC-CS502/2024-25
2024
DISCRETE MATHEMATICS

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A [OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer *all* questions

1. Find the remainder when the sum $(1!+2!+3!+4!+\dots+100!)$ is divided by 11.
2. Find the generating function $\{1,0,0,1,0,0,1,0,0,1,\dots\}$.
3. What is Equivalence relation?
4. State Pigeonhole Principle. Explain with an example.
5. The relation $R = \{(x, y) \in W, x^2 + y^2 = 25\}$. Find domain and range of R.

2
2
2
2
2

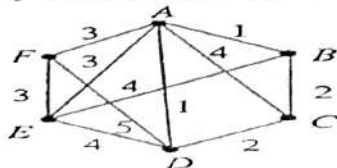
GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15 = 60

Answer any *four* questions

5+5+5

6. a Obtain the conjunctive normal form of $\{q \vee (p \wedge r)\} \wedge \{(p \vee r) \wedge q\}$.
- b Find the total number of integers between 1 and 3000 that are divisible by any of the integers 2, 5 or 7.
- c Show that $((p \wedge \sim q) \rightarrow r) \rightarrow (p \rightarrow (q \vee r))$ is a tautology.
7. a Verify the validity of the argument "If I drive to work then I will arrive in time. I do not drive to work. Therefore, I will not arrive in time".
- b State Euclid's Algorithm. Using this Algorithm find the G.C.D of (1120, 128).
- c How many ways the letters of the word MATHEMATICS be arranged in each of which the vowels occupy odd positions.
8. a Solve the recurrence relation $a_{n+2} - 4a_{n+1} + 4a_n = 3^n$, $n \geq 0$ with $a_0 = 2, a_1 = 1$ using generating function.
- b Find the missing digit in the number 8630480*874, if it is divisible by 11.
9. a Show that all roots of the equation $x^4 = 1$ form a commutative group under the operation multiplication.
- b Solve $u_{n+2} - 6u_{n+1} + 9u_n = (n+1)3^n$.
- c Using Prim's Algorithm find a minimal spanning tree from the following graph.



10. a If $f, g : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = ax + b$, $g(x) = 1 - x + x^2$ and $(go f) = 9x^2 - 9x + 3$, find the value of a and b . (5+5+5)
- b If $f(x) = [4 - (x-7)]^{1/5}$, find $f^{-1}(x)$.
- c Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. (5+5+5)
11. a Prove that a group (G, o) is abelian if and only if $(a o b)^{-1} = a^{-1} o b^{-1}$, for all $a, b \in G$.
- b Five dice are thrown simultaneously. In how many ways a total of 16 can be obtained?
- c Obtain DNF of $p \rightarrow (p \rightarrow q) \wedge \sim(\sim q \vee \sim p)$.
12. a State Hasse diagram. Draw the diagram for the PO set $(A, /)$ where $/$ stand for divisibility and $A = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$. (5+5+5)
- b Show that the following relation R on $\mathbb{Z}$ is an equivalence relation:
 $R = \{(a, b) : a, b \in \mathbb{Z} \text{ } a^2 + b^2 \text{ is a multiple of } 2\}$.
- c Using principle of mathematical induction prove that $4^{(2n+1)} + 3^{(n+2)}$ is divisible by 13.

Jalpaiguri Govt. Engg. College
(A Govt. Autonomous College)
COE/B.Tech./CSE/PCC-CS504 /2024-25
2024
OPERATING SYSTEM

FM: 70

Time Allotted: 3 hours

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

Group – A
[Objective Type Questions]

Answer **all** questions

5 X 2 = 10

1. What are disadvantages of multi-processor systems?
2. What are overlays?
3. Describe the system state when time quantum is zero in Round Robin scheduling.
4. "Switching is pure overhead to the system". Explain.
5. What is a kernel?

Group – B
[Long Answer Type Questions]

Answer any **four** of the following

4 X 15 = 60

6. a. What are the services provided by an operating system?
b. What is cooperating process? What are the reasons for providing environment of process cooperation?
c. Draw and explain the life cycle of a process.

5 + (2+3) + 5
7. a. What is Process Control Block? What is the information stored here?
b. Write short notes on Virtual memory?
c. What is processor affinity?

(2+5) + 5+3
8. a. What is race condition? Explain with an example.
b. What are the requirements for solution to the critical section problem?
c. Under which circumstances CPU-scheduling decisions takes place?
d. What is busy waiting? How it can be overcome?

5+3+3+(2+2)
9. a. What are the necessary and sufficient conditions for deadlock? Describe them.
b. What is semaphore? Why signal() and wait() are called atomic operations?
c. Consider a logical address space of 64 pages of 1024 words each, mapped onto a physical memory of 32 frames.
i. How many bits are there in the logical address?
ii. How many bits are there in the physical address?

5 + 5+ 5

PTO

10. a. Explain how a scheduling algorithm is called preemptive or non-preemptive with example.
b. What will happen in Round Robin scheduling if the time quantum is zero? Is the system will be in deadlock or not? Justify your answer.
c. What is thrashing and what is the cause of thrashing? 5+ 5+5
11. a. Why page size is always a power of two? What is demand paging?
b. If a system uses FIFO policy for page replacement, and it has 4-page frames with no pages loaded to begin with. Also, the system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order, then how many page faults will occur?
c. What is fragmentation? How it can be overcome? (2+3) + 5 + 5
12. a. What are the advantages of RAID level 3 over RAID level 1?
b. What are the differences between paging and segmentation?
c. When does a page fault occur? Explain various page replacement strategies/algorithms. 5 + 5 + 5

END

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/ CSE/ PCC-CS501/ 2024-25
2024
SOFTWARE ENGINEERING

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

5x2=10

- | | | |
|----|---|---|
| 1. | Write down the difference between Gantt chart and Pert chart. | 2 |
| 2. | What do you mean by Software Metrics? | 2 |
| 3. | Define Data Dictionary and its importance. | 2 |
| 4. | Write down the different types of Software Maintenance. | 2 |
| 5. | Write down the different types of System Testing. | 2 |

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions

4x15 = 60

- | | | |
|-----|---|-------------------|
| 6. | a) Draw a schematic diagram to represent the iterative waterfall model and explain the different phases.
b) Why Spiral model is called meta model? Identify three reasons of developing a prototype during software development.
c) What do you mean by size of a software project. Explain the different approaches to estimate project size. How COCOMO 2 differs from basic COCOMO? | 5
3+2
1+3+1 |
| 7. | a) What are the important types of risks that a project might suffer from? Explain the different stages of risk management.
b) Using Putnam's method show, how Effort varies with Development Time.
c) Write down the differences between project-based organization and function-based organization. Write down the possible ways that individual project teams are organized. | 3+4
3
3+2 |
| 8. | a) What do you mean by the terms cohesion and coupling in the context of software design?
b) Write down the significance of SRS in software development? What is the difference between functional and non-functional requirements?
c) According to Boehm's, explain different categories of software product with examples? Assume that the size of an utility software has been estimated to be 82,000 lines of source code. Assume that the average salary of software developers is Rs. 40,000 per month. Determine the effort required to develop the software, nominal development time, the no. of engineers required and the cost to develop the product. | 3
2+2
3+5 |
| 9. | a) What do you mean by reliability of a software product? Explain the following Reliability Metrics: ROCOF, MTTF, MTBF. What do you mean by Transient Failure and Cosmetic Failure.
b) Why is it important for software development organization to obtain ISO 9001 certification? Write down the different levels and their Key Process Areas in the context of SEI-CMM. | 1+6+2
2+4 |
| 10. | a) Explain Data Flow-based Testing principle. Write down the roles of Driver and Stub modules. Why Black box testing is also called Functional testing? Explain the different approaches available to design black-box test cases.
b) Distinguish between Function oriented design and Object oriented design. What is the difference between a sequence diagram and a collaboration diagram? In which context would you use each. | 3+2+1+3
3+3 |

11. a) Differentiate between Software verification and validation. Write down the different Labels of Testing and their goals. 3+3
b) Define Cyclomatic complexity of a program. Write a C function to compute the GCD of two integers and draw the Control Flow Graph(CFG). Also find the Cyclomatic complexity of the function. 3+5+1
12. Write short notes(*any three*) 5x3
a) Halstead's Metric
b) Integration Testing
c) SRS Document
d) Decision Table and Decision Tree
e) UML

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/CSE/PCC-CS503/2024-25
2024
DATABASE MANAGEMENT SYSTEMS

Times: 3 Hours

Full Marks: 70

*The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.*

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer **all** questions

1. State the responsibilities of a database administrator (DBA).
2. What is physical data independence? Give example.
3. What is data dictionary?
4. Define aggregate function and rename function with example.
5. Define extraneous attributes with example.

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15=60

Answer any **four** questions

6. a) Let the following relation schemas be given and write the following SQL/MySQL queries. 6

Employee(Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name, Job_id, Salary)
Department(Dept_no, Dept_name, Manager_id, Address).

- i) Display all the dept numbers available in emp and not in dept tables and vice-versa.
- ii) Display total salary spent for each job category.
- iii) Display lowest paid employee details under each manager.

- b) Consider the following three tables. 9

Students (sid, sname, rating, age)

Reserve (sid, cid, day)

Cycle (cid, cname, color)

Express the following queries in Relational Algebra and draw the query tree for the respective queries.

- i) Find students who've reserved a red or a green cycle.
- ii) Find students who've reserved a red and a green cycle.
- iii) Find the names of students who've reserved all cycles.

7. i) Explain 3 schema architecture with example and diagram. 5
ii) Define key attribute, composite attribute, derived attribute, multi-valued attribute and cardinality constraint with example. 5
iii) Construct an E-R diagram for a university registrar's office. 5
A university registrar office maintains data about the following details: a) courses, including course id, title, credits, syllabus, and pre-requisite; b) course-offerings, including course id, year, semester, dept, teacher and class room; c) students, including roll no, name, and program; d) teacher, including teacher id, name, dept and designation.

8. i) Define Boyce-Codd normal form. How does it differ from 3NF. Why it is considered a stronger form of 3NF. 5
 ii) Explain lossless join and dependency preservation property for relation decomposition. 3
 iii) Consider the following relation and functional dependency set. 7
 $R = \{A, B, C, D, E, F, G, H, I, J\}$
 $F = \{AB \rightarrow C, BD \rightarrow EF, AD \rightarrow GH, A \rightarrow I, H \rightarrow J\}$
 What is the key for R? Decompose the relation into 2NF and then 3NF relations.
9. i) State the Armstrong's axioms and prove union, decomposition and pseudo transitivity rule using the axioms. 5
 ii) Compute the closure of the following set F of functional dependencies $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ for relation schema $R = (A, B, C, D, E)$. List the candidate keys for R. 6
 iii) Using the given functional dependencies, compute the canonical cover of F. 4
10. i) Define different types of join operations with example. 5
 ii) State the SQL or MySQL query to define a view from supplier table using first three attributes. Explain the importance of view in DBMS. 5
 iii) Consider the given supplier table and briefly explain the impact of different anomalies in the given database. 5

Sname	Address	Mobile	Item	Price
X enterprise	9 Merchant Road	9133379388	Bread	30
X enterprise	9 Merchant Rd	9133379388	Butter	90
X enterprise	99 Merchant Road	9133379388	Cake	85
X Ent.	9 Merchant Road	9933379388	Biscuit	25
Y Suppliers	77 MK Road	9199979388	Bread	32

11. i) Explain the distinction between the terms serial schedule and serializable schedule. 4
 ii) Describe two phase locking protocol with suitable example. Justify how 2 Phase locking protocol is guaranteed to maintain conflict serializability. 5
 iii) Consider the partial schedule S1 given below. State the execution difficulties of the transactions in the present scenario. What action should be followed to make the data available for the proper execution of both transactions? 6

T1	T2
lock-X(B) read(B) $B := B - 100$ write(B)	
	lock-S(B) read(A) lock-S(B)
lock-X(A)	

12. i) List all possible sequences of states through which a transaction may pass. Explain why each state transition may occur. 5
- ii) Consider the following concurrent schedule. Assume initial value of A is 300 and B is 200. 5
Check whether the ACID property is satisfied for concurrent execution of transaction T3 and T4 in the schedule.

T3	T4
Read(A) $A := A - 100$	
	Read(A) $Temp := A * 0.2$ $A := A - Temp$ Write(A) Read(B)
Write(A) Read(B) $B := B + 100$ Write(B)	
	$B := B + Temp$ Write(B)

$A = 300$ $B = 200$
 $A = 300 - 100 = 200$
 $A = 200$
 $Temp = 200 * 0.2 = 40$
 $A = 200 - 40 = 160$
 $A = 160$
 $B = 200$
 $A = 160$
 $B = 200$
 $B = 200 + 100 = 300$
 $Write(A) = 160$
 $B = A + Temp = 160 + 40 = 200$
 $Write(B) = 200$

- iii) Define precedence graph. Consider the precedence graph of Fig. A. Is the corresponding schedule is conflict serializable? Explain your answer. If it is serializable, then show the acceptable topological ordering of the transactions in the given graph.

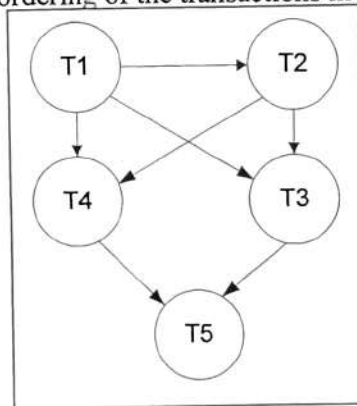


Fig.: A

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
 [A GOVERNMENT AUTONOMOUS COLLEGE]
 JGEC/B.TECH/CSE/PEC-CS501D/2024-25
 2024
 Operation Research

Full Marks: 70

Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer **all** questions

5x2=10

1. What do you mean by traffic intensity in a queue? 2
2. What is slack variable? 2
3. What do you mean by merge event in a network? 2
4. What do you do if rim condition is not satisfied in transportation problem? 2
5. Explain spanning tree with an example 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any **four** questions

4x15=60

6. i) Use the graphical method to solve the following Linear programming problem
 Maximize $Z = 8x_1 + 5x_2$
 Subject to constraints

$$2x_1 + x_2 \leq 500$$

$$x_1 \geq 150$$

$$x_2 \geq 250$$

$$x_1, x_2 \geq 0$$

- ii) Solve the following LPP using simplex method
 Maximize $Z = 3x_1 + 2x_2$
 Subject to constraints

$$x_1 + x_2 \leq 4$$

$$x_1 - x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

7. i) Solve the given transportation table using a) North-West corner) Least cost and c) VAM and calculate total transportation cost in each case.

(4+4+4)

	1	2	3	4	Availability
A	2	3	11	7	6
B	1	0	6	1	1
C	5	8	15	9	10
Requirement	7	5	3	2	

- ii) If the objective is to maximize the total profit for which the profit matrix is given, Write the approach to solve the problem.

3

8. i) By the notion of dominance, reduce the game and Solve the game graphically.

2+7

Player A	Player B		
	B1	B2	B3
A1	3	-3	4
A2	-1	1	-3
A3	1	-4	2

3)

1)

4)

- ii) Following Payoff Matrix is given

6

Player A	Player B*				
		B1	B2	B3	B4
	A1	-5	2	0	7
	A2	5	6	4	8
	A3	4	0	2	-3

Determine the best strategies and also the value of the game.
Is this game (i) fair (ii) strictly determinable?

9. i) The following table shows the jobs of a network along with their time estimates. The time estimates are in days: (3+6+2)

Job	1-2	1-6	2-3	2-4	3-5	4-5	5-8	6-7	7-8
t_o	3	2	6	2	5	3	1	3	4
t_m	6	5	12	5	11	6	4	9	19
t_p	15	14	30	8	17	15	7	27	28

- a) Draw the project network
b) find the critical path
c) Find the probability that the project is completed in 31 days (Given $P(Z \leq -2.1666) = 0.0114$)

- ii) State the similarities and distinction between PERT and CPM

4

10. i) Using the following cost matrix, determine (a) optimal job assignment (b) total cost of assignments. 10+1

Mechanic		JOB				
		1	2	3	4	5
	A	11	17	8	16	20
	B	9	7	12	6	15
	C	13	16	15	12	16
	D	21	24	17	28	26
	E	14	10	12	11	15

- ii) Write the differences between Transportation problem and Assignment Problem.

4

11. i) Use Big 'M' method to solve the following problem.

10

$$\text{Maximize } Z = x_1 - x_2 + 3x_3$$

subject to the constraints :

$$x_1 + x_2 \leq 20$$

$$x_1 + x_3 = 5$$

$$x_2 + x_3 \geq 10$$

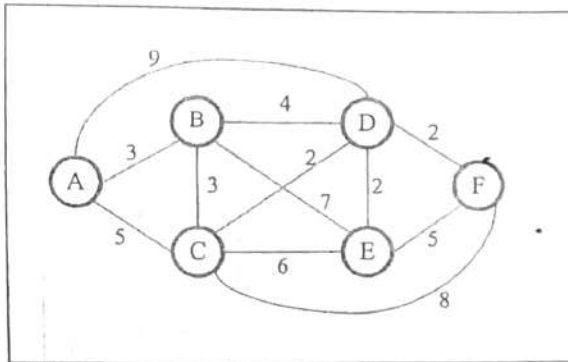
$$x_1, x_2, x_3 \geq 0$$

- ii) Classify Queueing models.

5

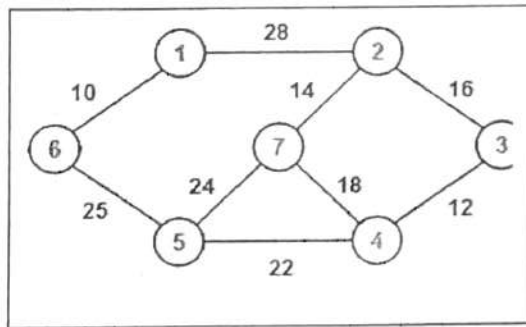
12. i) Using Dijkstra's algorithm find the length of the shortest path from source A to each vertex.

7



ii) Find the minimum spanning tree and calculate total duration using Kruskal's algorithm.

6



iii) Differentiate between Prim's and Kruskal's algorithm.

2

CE-1
①

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE

[A GOVERNMENT AUTONOMOUS COLLEGE]

JGEC/B.TECH/CSE/PEC-CS501B/2024-25

2024

SOFT COMPUTING

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A

[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

5x2=10

1. Define Fuzzy cardinality.
2. What is singleton?
3. What is Artificial Neural Network?
4. In what ways is bipolar representation better than binary representation?
5. Define bias and threshold.

2
2
2
2
2

GROUP-B

[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions

4x15=60

6. a. Let $U = \text{Fruits} = \{A, B, C, D, E, F\}$ be a universe on which two fuzzy sets, one of Ripped fruits and the other one of Fragrant Fruits are defined as shown below:
 $P = \text{Ripped Fruits} = 0.3/A + 0.9/B + 1.0/C + 0.7/D + 0.5/E + 0.4/F + 0.6/G$
 $Q = \text{Fragrant Fruits} = 1.0/A + 1.0/B + 0.5/C + 0.2/D + 0.2/E + 0.1/F + 0.4/G$
 Compute the Fuzzy sets $P \cup Q, P \cap Q, P', Q', P - Q, P \oplus Q$.
 b. What do you mean by Fuzzy if-then rule?
7. a. Define Fuzzy Inference System.
 b. Differentiate between crisp and fuzzy set.
 c. What do you mean by Generalized Modus ponens?
8. a. Find the max-min composition of the following fuzzy sets:

10
5
5
5
10

							z1	z2	z3	z4	
	y1	y2	y3	y4	y5		y1	0.9	0	0.3	0.4
R= $\begin{matrix} x1 \\ x2 \\ x3 \end{matrix}$	0.1	0.2	0	1	0.7	S= $\begin{matrix} y2 \\ y3 \\ y4 \\ y5 \end{matrix}$	0.2	1	0.8	0	
	0.3	0.5	0	0.2	1		0.8	0	0.7	1	
	0.8	0	1	0.4	0.3		0.4	0.2	0.3	0	
							0	1	0	0.8	

- b. Discuss about the various types of Soft Computing techniques.

5

SOFT COMPUTING

- | | | |
|-----|---|----|
| 9. | a. Discuss about different types of ' <i>Learning Techniques</i> ' in machine learning with example. | 5 |
| | b. What is activation function? Discuss about, Sigmoid function, Hyperbolic Tangent Function and the Rectifier linear unit with diagram. | 5 |
| | c. Define & differentiate "Hard vs. Soft Computing". | 5 |
| 10. | a. Develop the training algorithm to classify the logical 'AND' function using perceptron learning rule. Show a proper truth table of the training. | 10 |
| | b. Define perceptron learning rule. Also inform that in which way it is better than Hebb's learning rule? | 5 |
| 11. | a. Explain in detail the architecture of Mc Culloch – Pitts neuron model and also realize OR gate using the above neuron model. | 8 |
| | b. Show how multilayer perceptron overcome the classification problem of the patterns posed by the XOR problem. | 7 |
| 12. | a. With a neat flowchart, explain the simple form of Genetic Algorithm. | 8 |
| | b. What is the difference between Fuzziness vs. Probability? | 3 |
| | c. Show how individuals are selected from initial population by 'Roulette wheel selection' method. | 4 |

END